# Modular arithmetic

L3-NT1 Level 3 Senior · Number theory

Kevin Hu
November 2026

Use congruences, inverses, Fermat's little theorem and the Chinese remainder theorem to find remainders of very large numbers.
**By the end you can** compute remainders of huge powers using Fermat's little theorem and cycles; solve linear congruences by finding and using inverses; split a composite modulus and recombine with the Chinese remainder theorem.
**Prerequisites:** Factors, primes and divisibility (L2-NT1), Remainders and last digits (L2-NT2).
**Contests:** UKMT SMC, Senior Kangaroo, Maclaurin, AMC 12, AIME. **Time:** about 90 minutes.

## 1 Lesson

Consider the last two digits of $7^{2026}$, a number with 1713 digits. Note that $7^2 = 49$ and $7^4 = 2401$. The last two digits of $7^4$ are 01, so multiplying by $7^4$ never changes the last two digits of anything. Since $2026 = 4 \cdot 506 + 2$, we can write $7^{2026} = (7^4)^{506} \cdot 7^2$, and this ends in the same two digits as $1 \cdot 49$. The answer is 49.

Every step of that argument uses only remainders, and modular arithmetic is the notation that justifies discarding the rest of the number. This unit develops it: congruences, inverses, Fermat's little theorem and the Chinese remainder theorem.

### 1.1 Congruences

We write $a \equiv b \pmod{n}$, read "$a$ is congruent to $b$ modulo $n$", when $n$ divides $a - b$. Equivalently, $a$ and $b$ leave the same remainder on division by $n$. So $38 \equiv 3 \pmod{7}$, and also $38 \equiv -4 \pmod{7}$. Every integer is congruent to exactly one of $0, 1, \ldots, n-1$, but negative representatives are often the most useful ones: $99 \equiv -1 \pmod{100}$, and $2^{10} = 1024 \equiv -1 \pmod{25}$.

**Rules.**
If $a \equiv b$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d$, $a - c \equiv b - d$ and $ac \equiv bd \pmod{n}$. Hence $a^k \equiv b^k \pmod{n}$ for every positive integer $k$.

The first two are immediate. For the product, write $ac - bd = a(c - d) + d(a - b)$: both terms on the right are multiples of $n$. The power rule follows by using the product rule $k$ times. In practice this means that in any sum, product or power you may replace a number by anything congruent to it, at any stage. This is what we did with $7^4 \equiv 1 \pmod{100}$.

What you may *not* do in general is divide. We have $2 \cdot 3 \equiv 2 \cdot 8 \pmod{10}$, but $3 \not\equiv 8 \pmod{10}$. Division needs the next idea.



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## 1.2 Inverses and linear congruences

An *inverse* of $a$ modulo $n$ is a number $a'$ with $aa' \equiv 1 \pmod{n}$. For example $3 \cdot 5 = 15 \equiv 1 \pmod{7}$, so 5 is an inverse of 3 modulo 7. If $a$ has an inverse, then "dividing by $a$" means multiplying by $a'$.

**Theorem.**
The integer $a$ has an inverse modulo $n$ if and only if $\gcd(a, n) = 1$. In that case, for every $b$, the congruence $ax \equiv b \pmod{n}$ has exactly one solution modulo $n$.

*Proof.*  Suppose $d = \gcd(a, n) > 1$. If $ax \equiv 1 \pmod{n}$, then $ax - 1 = kn$ for some integer $k$; but $d$ divides both $ax$ and $kn$, so $d$ divides 1, which is absurd. Conversely, suppose $\gcd(a, n) = 1$ and look at the $n$ numbers $0, a, 2a, \ldots, (n-1)a$. If two of them were congruent, say $ia \equiv ja$ with $0 \le j < i \le n - 1$, then $n$ would divide $(i - j)a$. As $n$ shares no prime factor with $a$, unique factorisation forces $n \mid i - j$, which is impossible since $0 < i - j < n$. So these $n$ numbers leave $n$ different remainders, which means every remainder appears exactly once. In particular one of them is $\equiv 1$, and exactly one is $\equiv b$. ■

In a contest you rarely need Euclid's algorithm to find an inverse. Try small multiples, or better, look for a multiple of $a$ close to a multiple of $n$. Modulo 100, for instance, $33 \cdot 3 = 99 \equiv -1$, so $33 \cdot (-3) \equiv 1$ and the inverse of 33 is $-3 \equiv 97$.

**Example 1.**
Find the smallest positive integer $n$ such that $17n$ leaves remainder 5 when divided by 23.

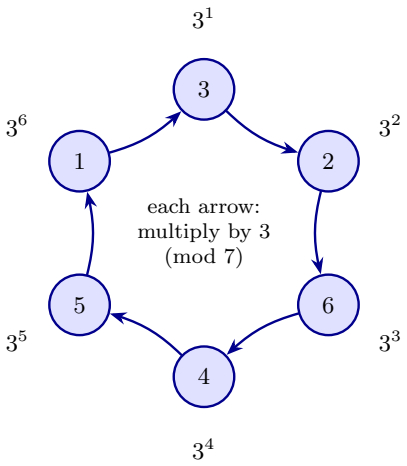
*Solution.*
We need $17n \equiv 5 \pmod{23}$. Testing $n = 1, 2, 3, \ldots$ happens to be quick here, but we use a method that also works for large moduli. First make the coefficients small: $17 \equiv -6$, so the congruence reads $-6n \equiv 5$, that is $6n \equiv -5 \pmod{23}$.
Now we want the right-hand side to be a multiple of 6, so keep adding 23: $-5 \equiv 18 \pmod{23}$, and $18 = 6 \cdot 3$. Since $\gcd(6, 23) = 1$, the number 6 is invertible modulo 23, so we may cancel it from $6n \equiv 6 \cdot 3$ and get $n \equiv 3 \pmod{23}$. The smallest positive answer is $n = 3$, and indeed $17 \cdot 3 = 51 = 2 \cdot 23 + 5$.
The method is to replace numbers by convenient representatives (17 by $-6$, then $-5$ by 18) until a common factor can be cancelled. The cancellation was valid only because 6 and 23 are coprime.

## 1.3 Fermat's little theorem

Consider the powers of 3 modulo 7: they are $3, 2, 6, 4, 5, 1$, and then the pattern repeats. We reach 1 after $p - 1 = 6$ steps, and the theorem below says that $a^{p-1} \equiv 1$ for every $a$ not divisible by $p$.



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**Fermat's little theorem.**
If $p$ is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Multiplying by $a$, every integer $a$ satisfies $a^p \equiv a \pmod{p}$.

*Proof.*  By the proof of the theorem on inverses (with $n = p$), the numbers $0, a, 2a, \ldots, (p-1)a$ are pairwise incongruent modulo $p$. Only the first is $\equiv 0$, so $a, 2a, \ldots, (p-1)a$ are congruent to $1, 2, \ldots, p-1$ in some order. Multiplying them all together gives $a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}$. Each of $1, 2, \ldots, p-1$ is coprime to $p$, so $(p-1)!$ is coprime to $p$ and can be cancelled, leaving $a^{p-1} \equiv 1$. ■

In practice this means that modulo a prime $p$, and for $p \nmid a$, exponents can be reduced modulo $p-1$. For instance, modulo 11 we have $a^{10} \equiv 1$, so $a^{2026} = (a^{10})^{202} \cdot a^6 \equiv a^6$. Sometimes the cycle is shorter than $p-1$ (the powers of 2 modulo 7 go $2, 4, 1$), but its length always divides $p-1$, as you will prove in L4-NT1. For a composite modulus like 100 there is no $p-1$ to use: either find the cycle directly, as we did for 7, or split the modulus, as below.

**Example 2.**
Find the remainder when $3^{2026}$ is divided by 31.

*Solution.*
Since 31 is prime and does not divide 3, Fermat gives $3^{30} \equiv 1 \pmod{31}$. Now $2026 = 30 \cdot 67 + 16$, so $3^{2026} \equiv 3^{16}$. This is still a large power, so we look for a small power of 3 close to a multiple of 31. First $3^3 = 27 \equiv -4$, which is small but not obviously useful. Then $3^5 = 243 = 7 \cdot 31 + 26$, so $3^5 \equiv 26 \equiv -5$.
Now $3^{15} = (3^5)^3 \equiv (-5)^3 = -125$, and $125 = 4 \cdot 31 + 1$, so $-125 \equiv -1$. Hence $3^{15} \equiv -1$, and $3^{16} \equiv -3 \equiv 28 \pmod{31}$. The remainder is 28.
Note the two habits: reduce the exponent first using Fermat, then keep the base small using negative representatives, so that no number in the calculation ever gets big.

## 1.4 The Chinese remainder theorem

Knowing a number modulo 4 and modulo 25 tells you the number modulo 100. More generally:



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**Chinese remainder theorem (CRT).**
If $\gcd(m,n) = 1$, then for any integers $a$ and $b$ the system $x \equiv a \pmod{m}$, $x \equiv b \pmod{n}$ has a solution, and any two solutions are congruent modulo $mn$.

*Proof.*  The $n$ numbers $a, a+m, a+2m, \ldots, a+(n-1)m$ all satisfy the first congruence. If two of them were congruent modulo $n$, then $n$ would divide $(i-j)m$ for some $0 < i-j < n$, which is impossible since $\gcd(m,n) = 1$, exactly as in the theorem on inverses. So they leave $n$ different remainders modulo $n$, and exactly one of them is $\equiv b$. If $x$ and $y$ are both solutions, then $m \mid x-y$ and $n \mid x-y$, and since $m$ and $n$ are coprime, $mn \mid x-y$. ■

So each pair (remainder mod $m$, remainder mod $n$) corresponds to exactly one remainder mod $mn$. The grid below shows this for 3 and 5: each of $0, 1, \ldots, 14$ sits in its own cell, and every cell is used once. The shaded cell says that the only number in this range that is 2 mod 3 and 4 mod 5 is 14.

| $n \bmod 3$ \ $n \bmod 5$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 0 | 0 | 6 | 12 | 3 | 9 |
| 1 | 10 | 1 | 7 | 13 | 4 |
| 2 | 5 | 11 | 2 | 8 | 14 |

The theorem extends to any number of pairwise coprime moduli. The coprimality matters: the system $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{6}$ has no solution at all, because the first condition makes $x$ odd and the second makes it even.

CRT is how you handle composite moduli. To work modulo 100, work modulo 4 and modulo 25 separately; for 1000, use 8 and 125. The pieces are usually much easier, especially when the base is not coprime to the full modulus.

**Example 3.**
Find the last two digits of $2^{2026}$.

*Solution.*
One might look for a power of 2 that is $\equiv 1 \pmod{100}$, as we did with 7, but there is none: every power $2^k$ with $k \geq 1$ is even, and an even number is never 1 more than a multiple of 100. The powers of 2 do eventually cycle modulo 100, but the cycle does not pass through 1, and finding it by hand is slow.
So split $100 = 4 \cdot 25$. Modulo 4 the question is trivial: $2^{2026} \equiv 0$. Modulo 25, the number 2 is invertible, and $2^{10} = 1024 \equiv -1$. So $2^{20} \equiv 1 \pmod{25}$, and since $2026 = 20 \cdot 101 + 6$, we get $2^{2026} \equiv 2^6 = 64 \equiv 14 \pmod{25}$.
Now combine. We want $0 \leq x < 100$ with $x \equiv 14 \pmod{25}$ and $x \equiv 0 \pmod{4}$. The first condition gives the candidates $14, 39, 64, 89$, and only 64 is a multiple of 4. So $2^{2026}$ ends in 64.

**Example 4.**
Find the remainder when $7^{7^7}$ is divided by 33. (The exponent is $7^7 = 823543$.)



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*Solution.*
The modulus is composite, so split $33 = 3 \cdot 11$. Modulo 3 there is nothing to do: $7 \equiv 1$, so $7^{7^7} \equiv 1 \pmod 3$.
Modulo 11, Fermat gives $7^{10} \equiv 1$, so all we need is the exponent $7^7$ modulo 10. This is the general method for towers: *to reduce the top exponent modulo $p-1$, you need a second, smaller modular calculation*. The last digits of the powers of 7 cycle $7, 9, 3, 1$ with period 4, and $7 = 4 + 3$, so $7^7$ ends in 3. Therefore $7^7 = 10k + 3$ for some integer $k$, and

$$7^{7^7} = \left(7^{10}\right)^k \cdot 7^3 \equiv 7^3 = 343 \pmod{11}.$$

As $343 = 31 \cdot 11 + 2$, this is 2.
Finally solve $x \equiv 1 \pmod 3$ and $x \equiv 2 \pmod{11}$. The candidates for the second are $2, 13, 24$, and only 13 is 1 more than a multiple of 3. The remainder is 13.
The tempting mistake is to reduce the exponent $7^7$ modulo 33, or modulo 11. Exponents are reduced modulo the length of the cycle, never modulo the original modulus.

**Common traps.**

- Cancelling a factor that shares a divisor with the modulus. From $6x \equiv 6 \pmod 9$ you may conclude only that $x \equiv 1 \pmod 3$, so $x \equiv 1, 4$ or $7 \pmod 9$.
- Reducing an exponent modulo $p$ instead of $p-1$, or modulo $n$ itself for a composite $n$.
- Using Fermat when $p \mid a$ (then $a^{p-1} \equiv 0$), or with a composite modulus: $2^8 = 256 \equiv 4 \pmod 9$, not 1.
- Applying CRT to moduli that are not coprime. Conditions mod 4 and mod 6 can contradict each other, and when they agree they pin $x$ down only modulo 12.
- Leaving the answer as $-3$ when the question wants a remainder between 0 and $n-1$.

### 1.5 How to spot this in a contest

The usual signals are "the remainder when ... is divided by", "the last two (or three) digits", "the units digit", a huge exponent, a year number such as 2026 in an exponent, or "for which $n$ is ... divisible by ...". When you see one, ask these questions in order.

1. Is the modulus prime? Then reduce exponents modulo $p-1$ using Fermat.
2. Is it composite? Split it into coprime prime powers and recombine with CRT: 4 and 25 for 100, 8 and 125 for 1000.
3. Is there a small power equal to $\pm 1$? Hunting for $\pm 1$, like $2^{10} \equiv -1 \pmod{25}$, is often faster than any theorem.
4. Is the exponent itself a power? Then compute the exponent modulo the cycle length first.

In multiple-choice papers, remember the cheap checks too: work mod 9 with digit sums, mod 11 with alternating digit sums, and mod 10 with last digits. A single small modulus often eliminates three or four options before you do any real work.



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## 2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** ⋆ Find the remainder when $2026^{2026}$ is divided by 7.

**Problem 2** ⋆⋆ Find the smallest positive integer $n$ such that the last two digits of $37n$ are 01.

**Problem 3** ⋆⋆ A positive integer $N$ leaves remainder 3 when divided by 7, remainder 4 when divided by 9, and remainder 5 when divided by 11. Find the smallest possible value of $N$.

**Problem 4** ⋆⋆⋆ How many integers $x$ with $1 \leq x \leq 1001$ have the property that $x^3 - x$ is divisible by 1001?

**Problem 5** ⋆⋆⋆ Find the remainder when $2^{3^{2026}}$ is divided by 13. (The exponent is $3^{2026}$.)

**Problem 6** ⋆⋆⋆⋆ Find the remainder when $1^{2026} + 2^{2026} + 3^{2026} + \cdots + 2026^{2026}$ is divided by 13.

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 10B 2016 Problem 8** ⋆
   *Only the last two digits matter: work mod 100.*
2. **AMC 12A 2008 Problem 15** ⋆⋆
   *Units digits of powers cycle; find the exponent modulo the cycle length.*
3. **AIME I 2017 Problem 2** ⋆⋆
   *Equal remainders means the divisor divides the differences.*
4. **UKMT SMC 2026 Question 8** ⋆⋆
   *Start from the largest candidate and track its remainder mod 7 as you change it.*
5. **BMO1 2019/20 Problem 1** ⋆⋆
   *Look at all the numbers involved modulo a small prime.*
6. **AMC 12B 2017 Problem 19** ⋆⋆⋆
   *Split 45 into coprime factors, use digit tests, then recombine.*
7. **AMC 10B 2011 Problem 23** ⋆⋆⋆
   *Work mod 1000 and keep the base small.*
8. **AIME 1983 Problem 6** ⋆⋆⋆
   *Work mod 49 and write each base as 7 plus or minus 1.*
9. **BMO1 2008/09 Problem 4** ⋆⋆⋆
   *Reduce the numerator modulo the divisor by replacing n with a congruent value.*
10. **AIME I 2010 Problem 2** ⋆⋆⋆
    *Most of the factors are congruent to -1 mod 1000.*



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11. **BMO1 2016/17 Problem 6** ⋆⋆⋆⋆

*Turn the four divisibility conditions into statements about one common expression.*

12. **AIME I 2011 Problem 11** ⋆⋆⋆⋆

*Use CRT with 8 and 125, and find where the powers of 2 start repeating.*



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## 4 Hints

1. Reduce the base modulo 7 first, then use Fermat to shrink the exponent.
2. You need $37n \equiv 1 \pmod{100}$. Look for a multiple of 37 that is very close to a multiple of 100.
3. Each remainder is a little more than half of its divisor. What can you say about $2N$?
4. Factorise 1001 into primes and factorise $x^3 - x$. Then count solutions modulo each prime and combine.
5. Fermat tells you to find the exponent modulo 12. Be careful: 3 is not coprime to 12, so split 12 into 4 and 3.
6. First reduce the exponent modulo 12. Then show that $S = 1^{10} + 2^{10} + \cdots + 12^{10}$ is divisible by 13 by comparing $S$ with $2^{10}S$.

## 5 Solutions

### Problem 1

*Find the remainder when* $2026^{2026}$ *is divided by* 7.

Since $2023 = 7 \cdot 289$, we have $2026 \equiv 3 \pmod{7}$, so we need $3^{2026}$ modulo 7. As 7 is prime and does not divide 3, Fermat gives $3^6 \equiv 1 \pmod{7}$. Now $2026 = 6 \cdot 337 + 4$, so $3^{2026} \equiv 3^4 = 81 = 11 \cdot 7 + 4$. The remainder is 4.

**Answer:** 4

### Problem 2

*Find the smallest positive integer* $n$ *such that the last two digits of* $37n$ *are* 01.

We need $37n \equiv 1 \pmod{100}$, so $n$ is the inverse of 37 modulo 100, which exists and is unique modulo 100 because $\gcd(37, 100) = 1$. Note that $37 \cdot 27 = 999 \equiv -1 \pmod{100}$. Multiplying by $-1$, we get $37 \cdot (-27) \equiv 1$, so $n \equiv -27 \equiv 73 \pmod{100}$. The smallest positive value is $n = 73$, and indeed $37 \cdot 73 = 2701$.

**Answer:** 73

### Problem 3

*A positive integer* $N$ *leaves remainder* 3 *when divided by* 7, *remainder* 4 *when divided by* 9, *and remainder* 5 *when divided by* 11. *Find the smallest possible value of* $N$.

Doubling the three conditions gives $2N \equiv 6 \equiv -1 \pmod{7}$, $2N \equiv 8 \equiv -1 \pmod{9}$ and $2N \equiv 10 \equiv -1 \pmod{11}$. So $2N + 1$ is divisible by 7, 9 and 11, and since these are pairwise coprime, it is divisible by $7 \cdot 9 \cdot 11 = 693$. Conversely, if $693 \mid 2N + 1$ then the three original conditions hold, because 2 is invertible modulo 7, 9 and 11 and the steps can be reversed. Now $2N + 1$ is odd, so it is an odd multiple of 693, and the smallest is 693 itself, giving $N = 346$. Check: $346 = 49 \cdot 7 + 3 = 38 \cdot 9 + 4 = 31 \cdot 11 + 5$.

**Answer:** 346



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### Problem 4

*How many integers $x$ with $1 \le x \le 1001$ have the property that $x^3 - x$ is divisible by 1001?*

We have $1001 = 7 \cdot 11 \cdot 13$ and $x^3 - x = (x-1)x(x+1)$. For a prime $p$, the prime $p$ divides this product if and only if it divides one of the three factors, that is, if and only if $x \equiv 0, 1$ or $-1 \pmod{p}$. For $p = 7, 11, 13$ these are three different residues, so there are exactly 3 good residues modulo each prime.

Since 7, 11 and 13 are pairwise coprime, $1001 \mid x^3 - x$ exactly when each of the three primes divides it. By the Chinese remainder theorem, each choice of a good residue modulo 7, modulo 11 and modulo 13 corresponds to exactly one residue modulo 1001, so there are $3 \cdot 3 \cdot 3 = 27$ good residues modulo 1001. The integers $1, 2, \ldots, 1001$ contain each residue modulo 1001 exactly once, so the answer is 27.

**Answer:** 27

### Problem 5

*Find the remainder when $2^{3^{2026}}$ is divided by 13. (The exponent is $3^{2026}$.)*

Since 13 is prime, $2^{12} \equiv 1 \pmod{13}$, so we need $3^{2026}$ modulo 12. We cannot use Fermat-style reasoning modulo 12 directly, since 12 is not prime and 3 is not coprime to it, so use the Chinese remainder theorem. Modulo 3, clearly $3^{2026} \equiv 0$. Modulo 4, $3 \equiv -1$ and 2026 is even, so $3^{2026} \equiv 1$. The residue modulo 12 that is 0 mod 3 and 1 mod 4 is 9, so $3^{2026} = 12k + 9$ for some integer $k$.

Hence $2^{3^{2026}} = (2^{12})^k \cdot 2^9 \equiv 2^9 = 512 \pmod{13}$, and $512 = 39 \cdot 13 + 5$. The remainder is 5.

**Answer:** 5

### Problem 6

*Find the remainder when $1^{2026} + 2^{2026} + 3^{2026} + \cdots + 2026^{2026}$ is divided by 13.*

Terms with $13 \mid k$ contribute 0. For $13 \nmid k$, Fermat gives $k^{12} \equiv 1$, and $2026 = 12 \cdot 168 + 10$, so $k^{2026} \equiv k^{10} \pmod{13}$. Since a term depends only on $k$ modulo 13, each block of thirteen consecutive values of $k$ contributes $S = 1^{10} + 2^{10} + \cdots + 12^{10}$ modulo 13.

The hard step is to show that $13 \mid S$. As $r$ runs through $1, 2, \ldots, 12$, the numbers $2r$ run through all the non-zero residues modulo 13 in some order (because 2 is invertible modulo 13). Therefore $\sum (2r)^{10} \equiv \sum r^{10}$, that is, $2^{10} S \equiv S$, so $(2^{10} - 1)S \equiv 0 \pmod{13}$. But $2^{10} = 1024 = 78 \cdot 13 + 10$, so $2^{10} - 1 \equiv 9$, which is invertible modulo 13. Hence $S \equiv 0$.

Now $2026 = 13 \cdot 155 + 11$. The values $k = 1, \ldots, 2015$ form 155 complete blocks and contribute 0. The remaining values $k = 2016, \ldots, 2026$ are congruent to $1, \ldots, 11$, so they contribute $1^{10} + \cdots + 11^{10} = S - 12^{10} \equiv 0 - (-1)^{10} = -1$. The remainder is $13 - 1 = 12$.

**Answer:** 12